Determine the derivative f' of the function $f(x) = x^2 \cdot 5^x$.

Remember that you can ***always*** *leave notes in order to obtain points for your work even if your final answer is incorrect.*

Answer

$f'(x) = \quad 2x \cdot 5^x + x^2 \cdot 5^x \cdot 4x^3 + \ln(5)$ ✗ Wrong

Exercise

Consider the lines l and k given by the equations

$$\begin{aligned} l: \quad ay + x &= a \\ k: \quad 4y - x &= a \end{aligned}$$

where a is a number such that l and k have an intersection point.

Give the intersection point of l and k: Give your answer as coordinates $[\square, \square]$.

Answer

$$\left[\frac{4a - a^2}{4 + a}, \frac{2a}{4 + a}\right]$$

✓ Correct

Determine the range of the function $f(x) = e^{8\cdot x} + \frac{4 \cdot e^{2\cdot x}}{3} - 4 \cdot e^{2\cdot x}$.

Give your answer in one of the forms:

- $y > y_1 \wedge y < y_2$, or $y < y_1 \vee y > y_2$, or
- $y < y_1$, or $y > y_1$, or
- *none* if the range is empty, or
- *all* if the range consists of all numbers,

where y_1 and y_2 are exact numbers, and the symbols $<$ and $>$ can be replaced by $\leq$ and $\geq$ in the expressions above.

Answer

$y \geq -\frac{5}{3}$ ✓ Correct

Determine whether $f(x) = -3 \cdot x^3 - 6 \cdot x^2 + 7 \cdot x + 3$ is increasing or decreasing at its inflection point.

You can only score full points with a proper explanation.

Answer

$f'(x) = -(9x^2) - 12x + 7$

$f''(x) = -18x - 12$

inflection point $= -\frac{2}{3}$

The orbit of a point P is defined by the following parametric equation for $-2 \cdot \pi \leq t < 2 \cdot \pi$:

$$\begin{cases} x(t) = \cos(t) + \cos\left(\frac{1}{2} \cdot t\right) \\ y(t) = \sin(t) + \sin\left(\frac{1}{2} \cdot t\right) \end{cases}$$

The point P passes through the x-axis in the origin and passes through the x-axis three more times. Determine the values $t = t_1$, $t = t_2$ and $t = t_3$ for which P passes through the x-axis at another point than the origin.

Give your answer as a list $[t_1, t_2, t_3]$.

Answer

$[\square, \square, \square]$ ✗ Wrong

The polynomial $k(x) = x^3 = 41 \cdot x^2 + 399 \cdot x + 441$ has a root in $x = -1$. Consider the function

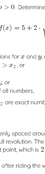
$$f(x) = \frac{x \cdot (2 \cdot x^2 - 58 \cdot x - 60)}{k(x)}$$

Give the **equation(s)** for the **vertical** asymptote(s) of $f(x)$.
If there is more than one, give all equations separated by the $\vee$ symbol (on the virtual keyboard).
If there are no vertical asymptotes, answer with *none*.

Answer

$y = -1 \vee y = 21$ ✗ One of the asymptotes is correct. The other is incorrect.

Let $f(x) = \frac{1}{x}$ and $a > 1$ be a real number. The points $P = [a, f(a)]$ and $Q = \left[\frac{1}{a}, f\left(\frac{1}{a}\right)\right]$ lie on the graph of $f(x)$. The origin O, P, and Q enclose a triangle in the plane. Determine the area of this triangle in terms of a.



Determine the antiderivative $F(x)$ of the function

$$f(x) = \sqrt{5 \cdot x + 3}$$

Consider the circle c with center $[0, 5]$ and radius 3. Furthermore, consider the family $\mathcal{F}$ of functions

$$f_a(x) = \frac{1}{a} x^2.$$

Determine the value(s) of a for which the circle c and the function $f_a(x)$ intersect twice.

Give your answer in the form of:

- *none*, if there are no values for a for which there are two points of intersection.
- $a = a_1$ if a_1 is the only value for a for which there are two points of intersection.
- $a = a_1 \vee a = a_2 \vee \ldots \vee a = a_n$ if there are n values $a_1, \ldots, a_n$ for a for which there are two points of intersection.

Remember that you can ***always*** *leave notes in order to obtain points for your work even if your final answer is incorrect.*

Hint: You can solve this question without determining the coordinates of the intersections. A sketch can help.

Consider the function $f_p(x) = \sqrt{p^2 - x^2}$, where $p > 0$.

Let l be the vertical line through the intersection of f_p with the x-axis. Determine the volume of the solid of revolution (around the x-axis) of the area determined by the region enclosed by the y-axis, the line l, the x-axis and the graph of f_p. *Please simplify your answer.*

Assume that p is a positive number, i.e. $p > 0$. Determine the domain and range of the function

$$f(x) = 5 + 2 \cdot \sqrt{3 \cdot x + \frac{x^2}{p}}$$

Give your answer in the form of expressions for x and y of the form

- $x > x_1 \wedge x < x_2$, or $x < x_1 \vee x > x_2$, or
- $x < x_1$, or $x > x_1$, or
- *none*, if the domain/range is empty, or
- *all* if the domain/range consists of all numbers,

where x can be replaced by y, x_1 and x_2 are exact numbers, and the symbols $<$ and $>$ can be replaced by $\leq$ and $\geq$ in the expressions above.

A Ferris wheel has 20 carts that are evenly spaced around the wheel. The wheel rotates at a constant speed and it takes 5 minutes to complete a full revolution. The total height of the Ferris wheel is 200 metres and customers enter the carts at the lowest point, which is 2 metres above the ground.

What is your distance from the ground after riding the wheel for 4 minutes and 10 seconds? Give an exact and fully simplified answer.

Consider the function

$$f_n(x) = \cos(x) - \sin(x - \alpha)$$

on the domain $[0, 2\pi]$.

For at least one value of α in $[0, 2\pi]$, the graph of $f_n(x)$ has an extreme value at $x = \frac{11 \cdot \pi}{6}$.

Calculate the extreme(s) value(s). Give your answer as $y = y_1$ or $y = y_1 \vee y = y_2 \cdots \vee y = y_n$, for the correct value(s) of $y_1, y_2, \cdots, y_n$.

Remember that you can ***always*** *leave notes in order to obtain points for your work even if your final answer is incorrect.*

The parametric curve

$$S: \begin{cases} x(t) &= t \cdot \cos(t) \\ y(t) &= t \cdot \sin(t) \end{cases}$$

forms a spiral in the plane. Prove that any line l: $y = a \cdot x$ through the origin intersects curve S at an angle unequal to 90° degrees (or $\frac{\pi}{2}$ radians).

You can only score full points with a proper explanation.

Hint 1 Do not try to find the specific values of t where S intersects l.

Hint 2 Either argue by using slopes of lines or by using direction vectors.
You may use that two vectors $\begin{pmatrix} a \\ b \end{pmatrix}$ and $\begin{pmatrix} c \\ d \end{pmatrix}$ are perpendicular when $ac + bd = 0$.

To get full points for this question, write your fully worked-out solution in the note field.

Consider the function $f(x)$ defined as

$$f(x) = \begin{cases} 3 - 2 \cdot x & \text{for } x < -3 \\ x^2 + 3 \cdot x + 9 & \text{for } -3 \leq x \leq 3 \\ 3 - 2 \cdot x & \text{for } x > 3 \end{cases}$$

Determine whether $f(x)$ has any jump discontinuities. Show in your notes for all candidates whether they indeed are a jump discontinuity or not.

Write down the x-coordinates of all jump discontinuities. Give your answer in the form of:

- *none*, if no jump discontinuity exists
- $x = x_1$ if only one jump discontinuity exists
- $x = x_1 \vee x = x_2 \vee \ldots$, if more than one jump discontinuity exists.

Write your fully worked out solution in the note field (click the Note button to open the field).

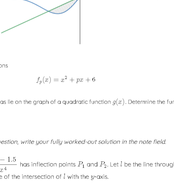
Consider the function

$$f(x) = x \cdot e^{-x^2}$$

and the linear function

$$g(x) = \frac{1}{e} \cdot x$$

As can be seen in the image below, the graphs of $f(x)$ and $g(x)$ enclose two regions of equal area. Determine the total area of these two regions, i.e. the sum of the two individual regions.



Consider the family of functions

$$f_p(x) = x^2 + px + 6$$

The vertices of these parabolas lie on the graph of a quadratic function $g(x)$. Determine the function rule of $g(x)$.

To get full points for this question, write your fully worked-out solution in the note field.

The function $f(x) = \frac{5x^2 - 1.5}{9x^4}$ has inflection points P_1 and P_2. Let l be the line through P_1 and P_2. Determine the y-coordinate of the intersection of l with the y-axis.

Don't forget to show that the points P_1 and P_2 that you find, are indeed inflection points.

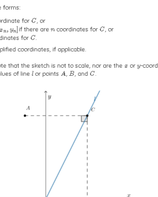
Write your fully worked out solution in the note field (click the Note button).

Determine the inverse of $f(x) = \log_5\left(\sqrt[3]{x}\right)$.

Consider the points $A = [-10, 2]$ and $B = [4, -6]$ and the line l described by $y = 5 \cdot x$. Find all points C on l such that $\angle ACB = \frac{\pi}{2}$.

Give your answer in one of the forms:

- $[x_1, y_1]$ if there is one coordinate for C, or
- $[x_1, y_1] \vee [x_2, y_2] \vee \ldots [x_n, y_n]$ if there are n coordinates for C, or
- *none* if there are no coordinates for C.

Write down the exact and simplified coordinates, if applicable.

See the figure for a sketch. Note that the sketch is not to scale, nor are the x or y-coordinates of any points specific to the actual values of line l or points A, B, and C.